**25900895** **A**

**Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

## FIRST SEMESTER

## Multi-Disciplinary Courses (MDC) - MG1MDCMAT100 - MATHEMATICS FOR COMPETITIVE EXAMINATIONS

(2024 ADMISSION ONWARDS)

**Duration: 1 Hours** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Multiple Choice Questions

Answer any **10** Questions

Each Question carries **2** marks

1. What is the place value of $9$ in the numeral $70984$ ? [E] / [CO1]

   a). $9$ b). $90$

   c). $900$ d). $9000$

2. The H.C.F of $2^3 \times 3^2 \times 5 \times 7^4, 2^2 \times 3^5 \times 5^2 \times 7^3, 2^3 \times 5^3 \times 7^2$ is [U] / [CO1]

   a). $980$ b). $680$

   c). $196$ d). $20$

3. Simplify $1148 \div 28 \times 1408 \div 32 =$? [U] / [CO1]

a). 1800 b). 1804

c). 1814 d). 1822

4. What is the cube root of $0.125$? [A] / [CO1]

a). $0.05$ b). $0.005$

c). $0.5$ d). $0.15$

5. The third proportional to$8$ and $12$ is [U] / [CO2]

a). $18$ b). $16$

c). $19$ d). $24$

6. How is $\frac{3}{4}$ expressed as percentage? [U] / [CO2]

a). $0.75\%$ b). $7.5\%$

c). $75\%$ d). $60\%$

7. Riya bought apples for ₹350. She sold them for ₹400. Find the percentage of profit made by Riya. [U] / [CO2]

a). $14.29\%$ b). $15.23\%$

c). $5.25\%$ d). $10.50\%$

8. What would be the simple interest earned on an amount of ₹$5760$ at the rate of 6 p.c.p.a. after 3 years? [A] / [CO3]

a). ₹$1036.80$ b). ₹$1063.80$

c). ₹$1336.80$ d). ₹$1666.80$

9. A jar contains black and white marbles. If there are ten marbles in the jar, then which of the following could not be the ratio of black to white marbles? [U] / [CO2]

a). $1:4$ b). $1:10$

c). $7:3$ d). $9:1$

10. No of odd days in 200 years ? [U] / [CO3]

a). 0 b). 1

c). 2 d). 3

11. Samir's age is one- fourth of his father's age and two third of his sister Reema's age. What is the ratio of the ages of Samir, Reema and their father respectively? [E] / [CO2]

a). 2:3:8 b). 1:2:3

c). 3:4:5 d). 5:7:9

12. In how many days can A, B and C together complete a work, if A can do the work alone in $12$ days, B alone in $10$ days and C alone in $6$ days? [K] / [CO4]

a). $\frac{21}{60}$ b). $\frac{20}{7}$

c). $\frac{7}{20}$ d). $\frac{60}{20}$

13. Find the cost of ₹ $6400$, $10\%$ stock at $15$ discount. [A] / [CO4]

a). ₹$5449$ b). ₹$5484$

c). ₹$5440$ d). ₹$5245$

14. In how many days will Rahul alone can complete a work, if Rahul is thrice as good as a workman as Arjun and together they finish that work in 24 days? [U] / [CO4]

a). 8 b). 18

c). 6 d). 32

15. A man walked at a speed of 4 km/hr from point A to B and came back from point B to A at the speed of 6 km/hr. What would be the ratio of the time taken by the man in walking from point A to B to that from point B to A? [A] / [CO4]

a). 2:3 b). 3:2

c). 3:4 d). 4:5

**[10x2 = 20]**

## Part B

Multiple Choice Questions

Answer any **10** Questions

Each Question carries **3** marks

16. The L.C.M. of $0.63,\ 1.05$ and $2.1$ is [U] / [CO1]

a). $63$ b). $21$

c). $630$ d). $6.3$

17. If $x + y = 15$ and $xy = 56$, then what is the value of $x^2 + y^2$? [U] / [CO1]

a). 110 b). 113

c). 121 d). Cannot be determined

18. $0.213 \div 0.00213 = ?$ [U] / [CO1]

a). $1$ b). $10$

c). $100$ d). $0.1$

19. The rational number for the recurring decimal $0.125125\ldots$ is [K] / [CO1]

a). $\frac{125}{999}$ b). $\frac{125}{900}$

c). $\frac{125}{99}$ d). None of these

20. If A earns $33\frac{1}{3}\%$ more than B, how much percent does B earn less than A? [E] / [CO2]

a). $25\%$ b). $20\%$

c). $30\%$ d). None of these

21. Sub-triplicate ratio of $(a : b)$ is [K] / [CO2]

a). $(a^{1/3} : b^{1/3})$ b). $(a^{1/2} : b^{1/2})$

c). $(a^3 : b^3)$ d). None of the above

22. A grocer sells rice at a profit of $10\%$ and uses weights which are $20\%$ less than the market weight. The total gain earned by him will be? [U] / [CO2]

a). $30\%$ b). $35\%$

c). $37.5\%$ d). $39\%$

23. A vendor sells 10 clips for one rupee gaining thereby $40\%$. How many clips did he buy for one rupee? [E] / [CO2]

a). 11 b). 15

c). 12 d). 14

24. If the price of a commodity decrease by $R\%$ then what is the increase in consumption so as not to decrease the expenditure? [K] / [CO2]

a). $\left[\frac{R}{(100+R)} \times 100\right]\%$ b). $\left[\frac{(100+R)}{R} \times 100\right]\%$

c). $\left[\frac{R}{(100-R)} \times 100\right]\%$ d). $\left[\frac{(100-R)}{R} \times 100\right]\%$

25. What will be the compound interest on ₹$25,000$ for 2 years at $4\%$ per annum? [A] / [CO3]

a). ₹$2050$ b). ₹$2100$

c). ₹$2200$ d). ₹$2040$

26. The ages of two persons differ by 16 years. If 6 years ago, the elder one be three times as old as the younger one, find their present ages of younger one. [E] / [CO2]

a). 18 years b). 16 years

c). 14 years d). 12 years

27. Which is better investment? $11\%$ stock at $143$ or $9\frac{3}{4}\%$stock at $117$. [A] / [CO4]

a). $11\%$ stock at $143$. b). $9\frac{3}{4}\%$stock at $117$.

c). Both are equally good. d). Cannot be compared.

28. The ratio of the speeds of a car, a train and a bus is 5 : 9 : 4. The average speed of the car, the bus and the train is 72 km/hr. What is the average speed of the car and the train together? [A] / [CO4]

a). 78 km/hr b). 82 km/hr

c). 84 km/hr d). 80 km/hr

29. A man covers a certain distance on a toy train. Had the train moved 4 km/hr faster, it would have taken 30 minutes less. If it moved 2 km/hr slower, it would have taken 20 minutes more. Find the distance. [A] / [CO4]

a). 50 km b). 60 km

c). 40 km d). 55 km

30. 45 men can complete a work in 16 days.Six days later they start working, 30 men joined them. How many time will now take to complete the remaining work? [U] / [CO4]

a). 27 b). 30

c). 7 d). 6

**[10x3 = 30]**

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